How-To Library

Informatica

Using the Snowflake advanced scanner in Data Privacy Management

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Abstract

You can configure a Snowflake advanced scanner in MetaDex to import and view data discovery results and to run subject registry scans in Data Privacy Management. This article explains the steps to run the configuration in MetaDex, import the resource from the catalog into Data Privacy Management, and to run a subject registry scan on the data store in Data Privacy Management.

Supported Versions

- Data Privacy Management 10.5.2.1

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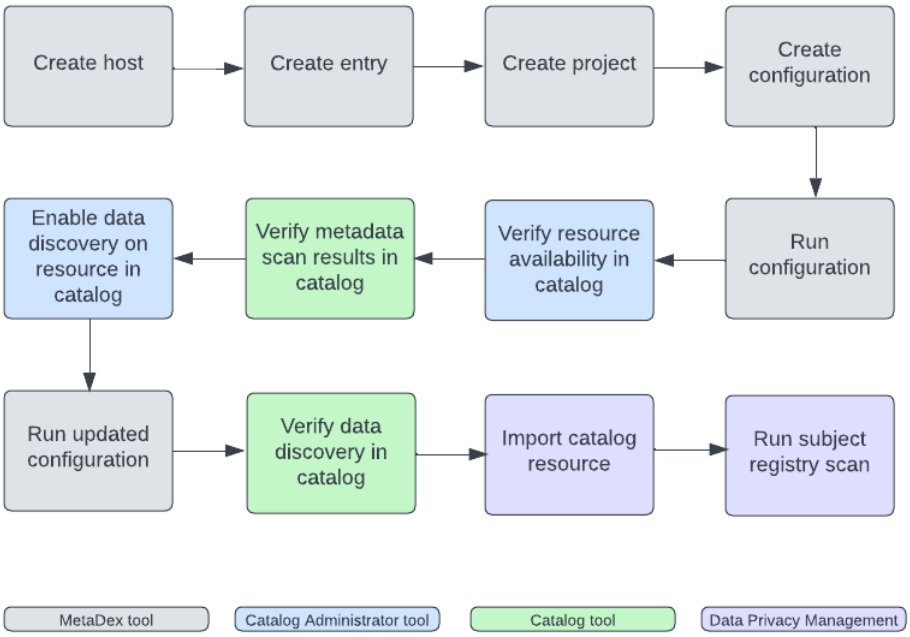
Overview

You can configure a Snowflake advanced scanner to run subject registry scans in Data Privacy Management. If you use the MetaDex tool to run data discovery on Snowflake resources in the catalog, you can import and view the results in Data Privacy Management.

You cannot create a Snowflake Advanced Scanner data store or run domain discovery scans directly in Data Privacy Management. You create a configuration in the MetaDex tool and update the resource metadata load settings in the Catalog Administrator tool. MetaDex runs the configuration and exports the results to the Catalog Administrator tool.

You then import the catalog resource into Data Privacy Management to view the data store details and domain discovery scan results. After you import the data store, you can then run subject registry scans on the data store in Data Privacy Management. You can use the Snowflake Advanced Scanner data store as a golden data store.

The following image shows the sequence of tasks to perform and indicates the tools used to perform each task:



Ensure that you perform all prerequisite tasks before you run scans with the Snowflake scanner.

Perform the following high-level tasks to run scans with the Snowflake scanner:

1. Create and run a configuration in the MetaDex tool. The resource becomes available in the Catalog Administrator tool. Enable data discovery for the resource in the Catalog Administrator tool and run the configuration again in the MetaDex tool. The data discovery results are imported and visible in the Catalog tool.

   Note: If you previously ran a configuration and can view the data discovery results in the Catalog tool, you don't need to run the configuration again.
2. Import the resource from the catalog as a data store in Data Privacy Management. You can edit specific properties of the data store after you import it, but you cannot import or export the data store from Data Privacy Management.
3. Run a subject registry scan in Data Privacy Management.

If you manually update the sensitive field information after you import the data store, the changes update in the Catalog Administrator tool. Repeat scans do not overwrite information that you manually update.

Prerequisites

Before you perform tasks in the MetaDex tool or the Catalog Administrator tool, ensure that you complete the following prerequisite tasks:

Note: To use the Snowflake scanner, you require the Enterprise Data Catalog and Advanced Scanners licenses in addition to the Data Privacy Management license.

- Ensure that you configured the MetaDex repository. See the *Enterprise Data Catalog Installation Guide* for information.
- Copy the required JAR and license files and restart the MetaDex tool.

- Create a connection to the Snowflake database in Informatica Administrator.

Copy the Snowflake JAR files and other required files

The Snowflake JAR files and other files must be available in the required locations before you perform tasks in the MetaDex tool.

1. Download the snowflake-jdbc-3.9.2 JAR file from the download location specified in Snowflake documentation.
2. Copy the JAR file to the following location: `<Data Privacy Management installation directory>/AdvScannersWorkspace/lib/jdbc`

   If the folder doesn't exist, create the folder and then copy the file.
3. Copy the MetaDex license file to the following location: `<Data Privacy Management installation directory>/AdvScannersWorkspace/etc`
4. Log in to the MetaDex tool.
5. Click **Administration** > **Global Variables**.
6. Click **Add variable** and add the following global variable and value: EDC_LICENSE_MODE=FILE
7. Click **Create**.
8. Run the following command to restart the MetaDex server:

```
sh $INFA_HOME/services/CatalogService/AdvancedScannersApplication/app/server.sh stop
nohup sh $INFA_HOME/services/CatalogService/AdvancedScannersApplication/app/server.sh &
```

Create a connection to the Snowflake database in Informatica Administrator

To scan a resource configured for data discovery and to import resources from the Catalog Administrator tool into Data Privacy Management, the Data Integration Service requires an Event Details Record (EDR) connection to the database. Create a connection to the Snowflake database in the Administrator tool.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.
3. In the Navigator, select the domain.
4. In the Navigator, click **Actions** > **New** > **Connection**.

   The **New Connection** dialog box appears.
5. In the **New Connection** dialog box, from the list of databases, select Snowflake and click **OK**.

   The **New Connection** wizard appears.

6. Enter the following connection properties:

Property	Description
Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters:~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. The ID must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	Optional. The description of the connection. The description cannot exceed 4,000 characters.
User name	The user name to connect to the Snowflake account.
Password	The password to connect to the Snowflake account.
Account	The name of the Snowflake account.
Warehouse	The Snowflake warehouse name.
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example:
	`user=jon&warehouse=mywh&db=mydb&schema=public`
	To access Snowflake through Okta SSO authentication, enter the web-based IdP implementing SAML 2.0 protocol in the following format:
	`authenticator=https://<your Okta account name>.okta.com`
	Note: Microsoft ADFS is not supported.
	For more information about configuring Okta authentication, see the following website: https://docs.snowflake.net/manuals/user-guide/admin-security-fed-auth-configure-snowflake.html#configuring-snowflake-to-use-federated-authentication

7. When you finish entering connection properties, you can click **Test Connection** to test the connection.
8. Click **Finish**.

Step 1. Perform configuration tasks in the MetaDex tool and the Catalog Administrator tool

Before you can import the data store and data domain scan results from the catalog, configure and run the configuration in the MetaDex tool.

Note: If you already ran a configuration in the MetaDex tool and can view the data discovery results for the resource in the Catalog tool, you do not need to repeat these steps. You can skip to "Step 2. Import the catalog resource into Data Privacy Management" on page 10.

Perform the following tasks in the MetaDex tool and the Catalog Administrator tool to enable data discovery for the resource and run the scan:

1. Create a host and entry in the MetaDex tool. A host and entry are similar to a data store group and data store in Data Privacy Management.
2. Create a project and configuration in the MetaDex tool. A configuration is similar to a scan in Data Privacy Management.
3. Verify that the resource and metadata scan results are available in the Catalog Administrator tool.
4. Edit the resource to enable data discovery in the Catalog Administrator tool.
5. Run the updated configuration in the MetaDex tool.

Step 1.1. Create a host and entry in the MetaDex tool

Create a host and entry to identify the database that you want to scan. A host is a group of databases and an entry refers to a specific database.

1. Log in to the MetaDex tool.
2. On the header, click **Administration**.
3. On the left pane, click **Connection Registry**.

   The Connection Registry Entries list appears on the right pane.
4. Click **Add host**.

   The **New host** window appears.
5. Enter a name for the host and click **Save**.

   The **Host** page opens.
6. Click **Add Entry** to add a database entry.

   The **New database** window appears.
7. Enter a database name.

   Enter the following configuration property details:
   - Repository type. Select JDBC from the list.
   - Vendor. Select Snowflake from the list.
   - Url. Enter the JDBC URL.
   - User. Enter the JDBC user name.
   - Password. Enter the JDBC password.
8. Click **Save**.

   The database page opens.

Step 1.2. Create a project and configuration in the MetaDex tool

Create a project to group scans that you run. A configuration specifies the scan settings. Create and run a configuration in the project to scan the required database.

1. Log in to the MetaDex tool.
2. On the **Projects** tab, click the **More** button, and select **Create project**.
3. On the **New project** page, enter a name and optional description for the project.
4. In the **Code** field, enter Snowflake as the database type.

5. Click **Save**.

   The Project page opens.
6. Click **Administration** > **Permission Management.**
7. Select the user that you use to run the tasks. On the user details page, click **Edit roles**.
8. On the **Projects** tab, click **Add** to add the project to the user.
9. On the **Roles** tab, drag the scroll bar to the right to assign Owner role.
10. Click **Save** to update the roles.
11. On the **Project** tab, click the **More** button and select **Create configuration**.
12. On the **New configuration** page, enter the following details:

Property	Description
Name	Enter a name for the configuration.
Description	Enter a description.
Parallel Safe	Select if you want to run multiple instances of the configuration simultaneously.
Conversion path	Select Advanced Database Objects to EDC.
Database object reader	From the **Read from** list, select the objects that you want to include in the scan. In the **Object include filters** field, enter the name of the database that you want to scan. Enter the name in the following format: `<host>.<database>.<schema>`

13. In the Enterprise Data Catalog section, enter the following details:

Property	Description
Output directory	Click the **Browse** button to select a folder from the **File Management** window. The scan stores the scan results file in the folder. The folder must be stored on the MetaDex server.
EDC address	Enter the Catalog service URL.
EDC user	Enter the Catalog Service user name.
EDC password	Enter the password.
EDC resource name	Enter the name of the resource to create in the Catalog Administrator tool.

14. Click **Save** to save the scan configuration.
15. Click **Run** to run the scan.

    The scan job page opens.

For more information about how to create and run a configuration in the MetaDex tool, see the *MetaDex Scanner Configuration Guide*.

Step 1.3. Verify that the resource and scan results are available in the catalog

When the MetaDex job completes, you can view the resource details and scan results. View the resource details in the Catalog Administrator tool, and view the scan results in the Catalog tool.

1. Log in to the Catalog Administrator tool.
2. Click **Library** > **Resources** and select the resource that you created in MetaDex.

   You can use the Resource Name filter on the Filter panel to quickly search for the resource. If the resource doesn't appear in the list, verify the status of the MetaDex job.
3. Select the resource to view the resource details. The resource type is Snowflake Advanced Scanner.

   You can monitor the status of the import of scan results from the **Monitoring** tab.
4. When the import job completes, log in to the Catalog tool.
5. Enter the name of the resource in the **Search** field and select the resource to open the resource page.
6. Click **Database** to view the schema that you selected in the MetaDex configuration.
7. Click **Tables** to view the list of tables in the schema.

   You can select a table to view the columns and other details on the **Asset** page.

The MetaDex scan job contains only metadata scan results. To scan the data, edit the resource in the Catalog Administrator tool and then run the configuration in the MetaDex tool.

Step 1.4. Enable data discovery in the Catalog Administrator tool

When the resource is available in the Catalog Administrator tool, update the properties required to enable data discovery on the resource.

1. Log in to the Catalog Administrator tool.
2. Click **Library** > **Resources** and select the resource that you created in MetaDex.

   You can use the Resource Name filter on the Filter panel to quickly search for the resource.
3. Select the resource to view the resource details.
4. Click the **Metadata Load Settings** tab.
5. Click **Edit** and update the following properties in the Data Discovery section:
   a. Select **Enable Data Discovery**.
   b. From the **Discovery Types**, select Profiling.
   c. Enter the following properties in the **Domain Connection Settings** section:

Property	Description
Specify the configuration settings for Data Integration Service	Choose Custom to enter the domain connection settings for the Data Integration Service.
Domain Name	Name of the domain.
Data Integration Service	Name of the Data Integration Service.
Username	User name to connect to the Data Integration Service.

Property	Description
Password	Password to connect to the Data Integration Service.
Security Domain	Name of the security domain.
Host	Host name for the Data Integration Service.
Port	Port number for the Data Integration Service.
Operating System Profile	Choose an operating system profile if the Data Integration Service is configured to use them. If you don't configure this, the Data Integration Service uses the default profile.

d. Enter the following properties in the **Common Discovery Options** section:

Property	Description
Source Connection Name	The name of the Snowflake connection that you created in Informatica Administrator.
Run On	Choose one of the following run-time environments: - Blaze. Runs the profile in the Hadoop environment on the Blaze engine. - Spark. Runs the profile in the Hadoop environment on the Spark engine. - Databricks. Runs the profile in the Databricks environment on the Databricks Spark engine in the Databricks cluster. - Native. Runs the profile on the same machine where the Data Integration Service runs.

e. Enter the following properties in the **Basic Profile Settings** section:

Property	Description
Profile Run Option	Choose one of the following profile types: - Column Profile. - Data Domain Discovery. - Column Profile and Data Domain Discovery.
Priority	Choose one of the following priority values: - High - Low
Sampling Option	Choose one of the following sampling options: - All rows. - Auto Random rows. - Random N rows. - First N rows. - Limit N rows. - Random Percentage

Property	Description
Exclude Views	Choose the option if you do not want the profiling scanner to scan the views in relational data sources.
Incremental Profiling	Choose the option to run the profile only on the changes made to the data source. If you do not select this option, the profile runs on the entire data source.
Cumulative	Choose this option to retain the column profile and column similarity results from the previous scan in the next resource scan results. If you do not choose this option, the previous profile results are purged.
Data Profile Filter	Enter the tables and views that you want to include or exclude in a profile run. The scanner extracts profile metadata based on the criteria that you specify.
Select Data Domain	Choose one of the following data domain options: - All Data Domains. - Specific Data Domain Groups. In the Data Domain Groups field, choose one or more data domain groups in the Data Domain Groups dialog box. - Specific Data Domains. In the Data Domains field, choose one or more data domains in the Data Domains dialog box.

6. Click Save.

You can now run the updated scan configuration from MetaDex.

For more information about how to update a resource in Enterprise Data Catalog, see the *Catalog Administrator Guide*.

Step 1.5. Run the configuration in the MetaDex tool

After you update the resource information in the Catalog Administrator tool, run the configuration from the MetaDex tool. The MetaDex server stores the results in the output directory and and then exports the results to the catalog.

1. Log in to the MetaDex tool.
2. Open the project and select the configuration.
3. On the **Configuration** page, click **Run**.

   You can monitor the status of the job from the **Jobs** tab on the page.

When the scan completes, the job exports the scan results to the catalog.

You can verify the data discovery results in the Catalog tool.

Step 2. Import the catalog resource into Data Privacy Management

When the configuration results are exported to the catalog as a resource, you can import the resource into Data Privacy Management as a data store.

Before you import catalog resources, use the satscurl command line utility to disable polling.

1. Download the Data Privacy Management satscurl command line utility from the following location in the Informatica tsftp server: `/updates/ILM/SecureAtSource/Tools/satscurl/10.5.2`

The following utility files are available for Windows and Linux operating systems:

- SATScurl_10.5.2_Windows.zip
- SATScurl_10.5.2_Linux.zip

2. Extract the files to the required location and run the following command: `./satscurl -a https://<host name>:<port number> -u <Data Privacy Management user name> -p <password> disablePolling`
   The host name and port number must be the values that you configure to access Data Privacy Management. For example: `./satscurl -a https://abccent7dsgace01.abc.com:18295 -u Administrator -p admin123 disablePolling`

1. Log in to Data Privacy Management.
2. Click **Manage** > **Data Stores**.

   The **Data Stores** workspace appears.
3. From the **Actions** menu, select **Import** > **Catalog Resources**.

   The **Import from Enterprise Data Catalog** page appears.
4. In the **Classification Policies** field, select the classification policies to evaluate in the data stores.

   Select the data domains or data domain groups that you used in the data domain scan.
5. Optionally, select the **Include row count** check box.
6. Optionally, clear the **Automatically update scan results after incremental scans** check box. Default is selected.
7. Select the catalog resource that you created.
8. Click **Import**.

   Data Privacy Management imports the data domain resources that match the selected classification policies for the selected data store. The data store appears in the list of data stores. The data store type appears as Snowflake Advanced Scanner.

You can view the scan results from the **Top Data Stores** widget on the **Overview** page. Use the Snowflake Advanced Scanner data store filter to search for the data store. Click the **Sensitive Fields** column value for the required data store to view the sensitive field information.

Run the following command to enable polling: `./satscurl -a https://<host name>:<port number> -u <Data Privacy Management user name> -p <password> enablePolling`

Snowflake Advanced Scanner data store properties

You can't create a Snowflake Advanced Scanner data store in Data Privacy Management, but you can view and modify the data store properties after you import the resource from the catalog. Properties that you modify in Data Privacy Management are updated in the Catalog Administrator tool. If you configure the data store to sync with the catalog, updates made to the data store in the catalog are visible in Data Privacy Management after the Sync Catalog job runs.

The following table describes the Snowflake Advanced Scanner connection properties:

Property	Description
Name	Required. Short description of the data store. The name is not case sensitive and must be unique within the Data Privacy Management repository. The name cannot exceed 255 characters or contain spaces or the following special characters: `\~!$%^&*()+`
Description	Optional. Long description of the data store that does not exceed 255 characters.
Category	Read only. Indicates the data store type as Cloud.

Property	Description
Data Store Type	Read only. Snowflake Advanced Scanner.
Account	Name of the Snowflake account. In the Snowflake URL, your account name is the first segment in the domain. For example, `123abc` is your account name in the URL `https://123abc.snowflakecomputing.com`.
User	The user name to connect to the Snowflake account.
Password	Password associated with the Snowflake user account.
Source Connection Name	Connection name in the Informatica domain that the Cloud job uses to connect to the Snowflake database.

You can configure the following additional properties:

Property	Description
Data Store Group	Required. Select a logical grouping of related data stores from the list of groups defined on the **Data Store Groups** workspace. You can assign a data store to one data store group.
Security Groups	Required. Select a security group from the list of all security groups in the Informatica domain, including native and LDAP groups. Native groups display the group name. For example, Administrator. LDAP groups include the security domain and the group name. For example, <*security domain*>/<*LDAP group name*>.
Tags	Optional. Keywords that add to the description. Create a new tag or select an existing tag from the list. You can assign multiple tags to a data store. After you assign a tag to a data store, you can use tags in alert rules to improve the specificity of the alert. For example, you want to be alerted when the risk score of any production data store increases by 20%. However, you do not want to be alerted for non-production data stores. You can assign the tag PROD to the production data stores. Then, configure the alert conditions to alert you when the risk score increases by 20% for data stores that have the PROD tag. When you delete a data store, if a tag was only assigned to the deleted data store, the Data Privacy Management Service deletes the tag from the repository.
Department	Optional. Enter the department that is responsible for the maintenance or ownership of the data store. The department can be a line of business, such as investment banking or wealth management, or a corporate department, such as Sales or Marketing. The name is case-sensitive and cannot exceed 100 characters. After you scan data stores, you can view a summary of scan results for all data stores by department. You can filter the scan results by location on the **Overview** workspace.
Data Owner	Optional. Select the individual, group, or organization name that is responsible for the accuracy and integrity of the source data on which you run a classification profiling scan.
Shared with Third Party	Optional. Indicates that the data in the data store is shared with a third party such as a third-party organization or vendor.
Third-Party Description	Optional. Appears if you select the **Shared with Third Party** check box. Enter a description of the third party. You can enter up to 2000 characters and use the following special characters: , _ @

Property	Description
Location	Optional. Select a location that is defined on the **Locations** workspace. If you do not select a location, Data Privacy Management assigns the Unknown location to the data store.
Auto Sync Catalog	Optional. Determines if the Scan job automatically synchronizes data source information in the catalog with the corresponding data store information in Data Privacy Management. Default is disabled.

The following read-only properties appear on the data store details page. The values do not impact the data store:

Property	Description
Auto Assign Connections	Select to automatically assign the schemas in Enterprise Data Catalog to the data store. Default is No.
Case Sensitive	Specifies that the data store is configured for case sensitivity. Select the check box to specify that the data store is case sensitive. Clear the check box to specify that the data store is case insensitive. Default is selected.
Memory	Memory that the Enterprise Data Catalog scanner uses when you run a scan on the data store. Select **High**, **Low**, or **Medium**. Default is Low. You might want to increase the scanner memory in the following example scenarios: you scan a large data store, you do not specify the schemas for a data store, or you scan multiple data stores in parallel. If you change the scanner memory, you must change the memory settings in the Yarn Resource Manager.
Custom JVM Options	JVM parameters that configure the scanner container. Configure the following parameters: - `Dscannerloglevel=<DEBUG/INFO/ERROR>`. Changes the log level of the scanner to values such as DEBUG, ERROR, or INFO. Default value is INFO. - `Dscanner.container.core=<Number of cores>`. Increases the core for the scanner container. The value must be a number. - `Dscanner.yarn.app.environment=<key value>`. Key pair value that you need to set in the YARN environment. Use a comma to separate the key pair values. - `Dscanner.pmem.enabled.container.memory.jvm.memory.ratio=<1.0/2.0>`. Increases the scanner container memory when `pmem` is enabled. Default value is 1.
Enable Reference Resources	Select to extract metadata about resources in Enterprise Data Catalog that the data store references. Default is No.
Retain Unresolved reference Assets	Appears when you select the Enable Reference Resources property. Reference assets are assets from reference resources in Enterprise Data Catalog, such as reference data sources and reference data sets. Select this option to keep unresolved reference assets. Default is Yes.

Step 3. Run a subject registry scan in Data Privacy Management

You can run subject registry scans on the Snowflake Advanced Scanner data store that you import from the catalog.

1. Click **Manage** > **Scans**.

   The **Scans** workspace appears.

2. Click **New**.

   The **Define & Connect** panel appears.

3. Enter a name and a description of the scan.

4. Choose the scan type. Select one or more of the following options:
   - **Discover Subjects**. Scans golden data stores to identify subjects.
   - **Link Subjects**. Links subjects in the data stores with subjects identified in golden data stores to ensure that the subject registry includes all subject data.
5. Select one or more subject types that appear in the list.
6. Select the Cloud category and Snowflake Advanced Scanner data store type.
7. Select the data store that you imported from the catalog.
8. Choose to configure repeat scans as incremental or full. Select the **Full Subject Scan** option to configure repeat subject registry scans as full scans. Clear the option to configure repeat scans as incremental. Default is selected.
9. If you selected **Link Subjects**, select the scan options.
10. Click **Next**.

    The **Assign Classification Policy** panel appears.
11. If you selected **Discover Subjects** only, you cannot assign classification policies. Go to step 10. Else select the classification policies to assign to the scan. You can use the following filters:
    - **Classification Policies Selected** . Displays a list of the classification policies that are assigned to the scan. Use the filter to view or remove classification policies.
    - **Assign Classification Policies**. Displays a list of all classification policies in the Data Privacy Management repository. Use the filter to add or remove classification policies.
12. Click **Next**.

    The **Schedule** panel appears.
13. Schedule the scan.

    You can schedule the job to run immediately, in the future, or on a recurring schedule.
14. Click **Save**.

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